Question			Answer	Marks	AO	Guidance	
12	(a)		Letting $h=0$ does not make sense as this has the two points coincident Dividing by zero is not allowed $\frac{0}{0}$ is not well defined $\frac{0}{0}$ is not equal to 1	**B1** **[1]**	**2.3**	Statement identifying a fault – should be equivalent to one of these statements Do not allow for $\frac{0}{0}=0$ or $\frac{0}{0}=6$ etc	
12	(b)		$\lim_{h\to 0}\left(\frac{f(3+h)-f(3)}{h}\right)$ gradient of chord is $6+h$ gradient of the curve is 6	**M1** **M1** **A1** **[3]**	**2.1** **2.1** **2.1**	replaces $h=0$ with the idea of limit for an expression which is not just $\lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)$ Allow for the value 3 used or this function ie $\lim_{h\to 0}\left(\frac{(x+h)^2-x^2}{h}\right)$ Simplifies the fraction Must be seen	
12	(c)		Gradient of the normal is $-\frac{1}{6}$ Equation of the normal is $y-9=-\frac{1}{6}(x-3)$	**M1** **A1** **[2]**	**1.1a** **1.1b**	Allow for $y=-\frac{1}{6}x+c$ seen even if their c is incorrect FT their (b) Any form FT their (b); ISW	Allow for a value found by an attempt to differentiate $y=x^2$ not from first principles $x+6y=57$